MILITARY SPECIFICATION SHEET

SHEET, PRINTED WIRING BOARD, PREPREG, BASE MATERIAL BI
(NONWOVEN ARAMID REINFORCEMENT, POLYIMIDE RESIN, FLAME RESISTANT)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-S-13949.

REQUIREMENTS:

Reinforcement material: B for nonwoven aramid. All parameters shall be as detailed in table I.

Resin system: I in accordance with MIL-S-13949.

Special consideration (third letter designator): J or L in accordance with MIL-S-13949.

Qualification inspection.

Inspection routine: The inspection routine shall include the permittivity and loss tangent tests in accordance with MIL-S-13949.

Extent of qualification:

Qualification to special consideration L shall extend to special consideration J.

Qualification of any reinforcement material style in table I shall extend to all other reinforcement material styles in table I.

Conditioning of specimens prior to test: Unless otherwise specified herein, specimens shall be conditioned in accordance with the applicable test method.

Property values, all reinforcement styles.

Flammability, duration of burning time, each flame application: Not applicable.

Volatile content.

Special consideration J (conditioning of specimen prior to test, E-0.25/163): 2.5 percent maximum.

Special consideration L (conditioning of specimen prior to test, E-0.5/225): 2.5 percent maximum.

Electric strength, average.

For base material thicknesses greater than .005 inch (0.13 mm) and less than .020 inch (0.51 mm)): 750 volts/mil minimum.

For base material thicknesses less than .005 inch (0.13 mm)): 500 volts/mil minimum.

Loss tangent (at 1 MHz): .035 maximum.

Permittivity (at 1 MHz): 4.5 maximum.

DISTRIBUTION STATEMENT A. Approved for public release, distribution is unlimited.

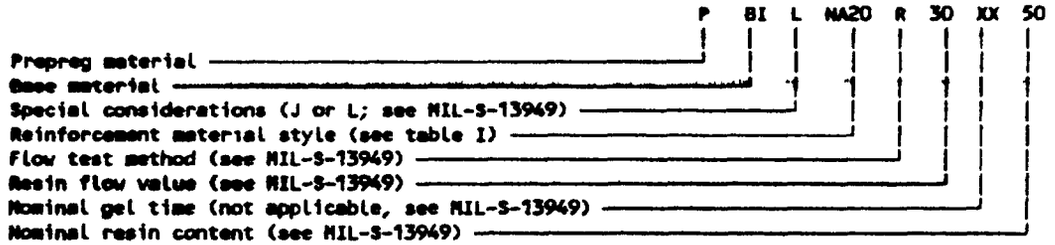
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TABLE I. Nonwoven aramid reinforcement construction and prepreg characteristics.

Reinforcement material		Prepreg selection data	
Style	Weight, ±15 percent (ounces per square foot)	Resin content range (percent)	Nominal thickness range, cured inches/(mm)
NA10	1.00	40 - 70	.0020 - .0050/(0.05 - 0.08)
NA16	1.60	40 - 70	.0025 - .0035/(0.06 - 0.09)
NA20	2.00	40 - 70	.0035 - .0050/(0.09 - 0.13)
NA25	2.50	40 - 70	.0040 - .0055/(0.10 - 0.14)

Supersession information: Not applicable.

Type designation: Consists of "PBI" prefix followed by coded alphas and numeric characters as shown in the following example:



CONCLUDING MATERIAL

Custodians
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - MI
Navy - MC
Air Force - 99

Preparing activity:
Army - ER

Agent.
DLA - ES

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